

KOHKI KITA

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EDUCATION

San Diego State University

San Diego, CA

B.S. in Computer Engineering

June 2026

Relevant Coursework: Data Structures (C++), Embedded Systems (C), Advanced Programming (Python), Microprocessors (Verilog), Windows Database and Web Programming (SQL, HTML)

WORK EXPERIENCE

Aztec Electric Racing

San Diego, CA

Controls Software Engineer

August 2024 - Present

- Enhanced battery management system software using ESP32 and the Arduino platform
- Developed and implemented a flag-based system to optimize cell balancing functionality
- Led the integration of CAN communication into the battery management system, enabling transmission of error and flag messages to the driver

Cyan Swim Academy

Kirkland, WA

Private Swim Instructor

March 2021 - August 2024

- Taught basic skills of swimming to over 30 kids
- Created comprehensive fitness programs tailored to the needs and goals of each individual
- Provided emergency first aid and CPR as needed

PROJECTS

FPGA Record Player

February 2025 - Present

Verilog, FPGA

- Implemented real-time audio processing using an FPGA
- Created interfacing with an ADC for digitization and a DAC for playback
- Designed memory management system using volatile and nonvolatile memory

AVR Microcontroller ADC with LED

November 2024

AVR Microcontroller, C

- Implemented ADC and timers to control LED brightness proportional to potentiometer input
- Configured dual timers for interrupt-driven ADC sampling and PWM generation at variable frequencies

Personal Portfolio Website

July 2024 - September 2024

HTML/CSS, JavaScript, React.js

- Created dynamic and interactive web application with React Framework
- Implemented seamless user experience across various devices with responsive design principles
- Achieved a clean and modern user interface with a focus on intuitive navigation and accessibility
- Integrated a contact form using Email.js to enable users to send messages directly from the website

IoT Wireless Image Display System

June 2024 - August 2024

ESP32, Arduino, C++

- Developed an IoT image display system using ESP32 with an LCD screen utilizing the SPI interface
- Integrated Adafruit libraries for efficient graphics rendering on the LCD

SKILLS & INTERESTS

Programming Languages: C, C++, Verilog, Python, JavaScript

Tools: Git, VS Code, Arduino, Figma, Adobe Lightroom

Languages: English, Japanese

Interests: Photography, Cooking, Lifting, Music, and Working on Cars